Measuring Player Fatigue to Optimize Individual Performance

SMT DATA CHALLENGE - Undergraduate Division

Team 69

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ABSTRACT

Professional baseball's marathon-esque conditions have only become more defined in the modern era of baseball, as players are now expected to perform high-intensity movements at higher rates. Resultantly, player fatigue concerns have created new discussions surrounding load management and player usage; organizations are now looking for ways to quantify physical exertion to gain an edge in the new sabermetric era. This study aims to offer a form of quantification about the issue, utilizing infield ground ball play trends to create a correlation-based case study to examine professional baseball promotion rates in relation to exertion and fatigue. By examining sprint speed and a computer-engineered, RPE-based fatigue metric, we sought to find a correlation between these metrics and the promotional pipeline in professional baseball. Despite the potential of our statistical analysis and engineered fatigue metric, the study found no correlation between the selected metrics and promotion rates. The study presents ramifications and initial steps toward integrating load management & fatigue systems to optimize the long-term value and performance of these professional baseball athletes.

INTRODUCTION & CONTEXTUALIZATION

Professional baseball, in its most pure state, is described by many as a marathon-esque environment throughout a season. The sport has only continued to evolve as high-intensity, explosive movements become more ingrained in the forefront of the player development pipeline. With all of this change, one issue has become apparent: the lack of coordination between the sport's data and physiology. In our exploration, we wanted to create some quantification of this volume of physical exertion within the data presented to us. Recent history has shown player usage at the professional level taking a drastic change for the worse. For pitchers, arm injuries are at an all-time high due to the increasing incentives for velocity, and according to the American Medical Association, 35.3% of active MLB pitchers had undergone Tommy John surgery in 2023, a 29% increase since 2016. Batters may not be putting the same level of stress on their bodies, but the sheer volume of plays contribute to wear and tear over time.

Load management is an idea that persists in the inner circles of baseball, from the amateur ranks, where high school arms have inning limits, to the professional ranks, where athletes' seasons are cut short due to the organization wanting to protect the individual athlete as an asset. In an ESPN article by Jesse Rogers, he states: “At a time when analytics have become a standard element of almost every front office decision, optimizing player workload is seen as one of the few remaining areas teams can gain an edge.” The article even cites the Milwaukee Brewers General Manager, Matt Arnold, stating: "Soccer and NBA teams have been tracking this kind of stuff for years. We have room to grow in our industry." Finding a way to quantify fatigue is a point of emphasis within the sport.

PROCESS

It is important to note that not all plays are equal when considering a player's exertion. For example, it is expected for a batter to round the bases at a more leisurely pace after hitting a home run, but stealing a base might require much greater input. Thus, analogous to the view that pitch leverage is more predictive of pitcher fatigue than pitch count, and perhaps more intuitive as well, player exertion is not simply dictated by factors such as distance covered.

In this project, the focus is on the batter's exertion on infield ground balls. This specific subset of plays was selected for a few reasons.

1. Volume of plays: For most 9-inning baseball games, 27 outs need to be recorded by each team, meaning that the total number of plays in a season is very large. As one of the three main ways to get outs, ground balls occur very frequently, giving a large sample size for analysis. Additionally, all different types of players will be involved in these plays, in contrast to stolen bases, for example, which occur far less often overall but more frequently among faster players.
2. Exertion level: Barring special circumstances, batters are expected to hustle on ground balls to try and beat the force out, so these plays generally have a higher level of exertion compared to routine fly balls and base hits. While there are cases of less competitive plays where a batter may run slower, those are in the minority.
3. Clear, consistent definition: For this project, infield ground balls are defined using the conditions listed below. This is not a perfect filter, as errors or deflections may lead to the inclusion of some extraneous plays, but it is a standard definition that matches well overall and leaves no ambiguity.
    a. All fielders to contact the ball are infielders.
    b. The ball hits the ground at some point during the play.

c. The maximum XY-distance the ball travels from home plate is less than 180 feet, and the maximum height is less than 20 feet.

The standardized definition for play selection also simplifies the analysis of the metadata, which is discussed in more detail in the attached code. At a high level, the data were collected by filtering relevant plays from the event codes, using ball position data to further reduce the number of included plays. Then, player position data for the batter was collected for each play, and it was used to compute the speed and exertion metrics to be analyzed.

The provided data do contain some issues that limit the viability of the extraction. The tracking itself was a bit imprecise, sometimes with large jumps in instantaneous speed, inaccurate renderings, and missing time intervals sprinkled throughout the data (see **Appendix A, Fig. 1**). At the cost of precisely computing player acceleration, we determined that the best approach to compute speed was averaging over a time interval to minimize the impact of individual points. By using the Statcast definition of sprint speed (speed over a one-second interval), the data were much more accurate to reality. Still, the presence of such inconsistencies represents a source of systematic error that should be noted when considering the results.

Aside from instantaneous and Statcast sprint speed measuring how fast the player is moving at a given time, we also considered an average speed metric. Similar to Home-to-First, defined as the time between contact with the ball and the batter reaching first base, average speed generalizes this to include plays where the batter position is not tracked through first base. This is another important metric that scouts have used for decades to characterize a player's speed, reaction time, and hustle, albeit at a broader level.

The question of whether maximum or average values should weigh more arises in this analysis. A player who has a very fast but inconsistent peak speed may not be as valuable as another player who is consistently fast but cannot reach as high of a peak. Thus, to highlight the importance of player recovery from play to play, both the maximum and average values were computed for the speed metrics that we considered. The distinction between maximum and average also forms the basis for the key idea of this project, measuring the amount of player fatigue and its impact on performance.

In a sports setting, it is fair to assume that a greater speed directly relates to a greater level of exertion by the athlete. Finding an exact relationship is complicated, likely requiring a biomechanical approach to player movement, but the general trend should not deviate significantly. Thus, our goal within this project was to develop a model for exertion from these metrics and analyze any correlation to promotion between levels. We used a fatigue metric with an RPE baseline based on information from Christine Luff in terms of running to quantify this measurement. The metric was simple in its construction, utilizing a cumulative methodology in which the feet traveled for each individual play would be multiplied by the RPE value that would be the percentile. The values are as displayed below:

Table 1. Fatigue factor numbers corresponding to each sprint speed percentile, computed at each frame for a play.

Fatigue Factor #	**Sprint Speed Percentile**
1	0%-52.5%
2	52.6%-55%
3	55.1%-57.5%
4	57.6%-60%
5	60.1%-70%
6	70.1%-75%
7	75.1%-80%
8	80.1%-87.5%

9	87.6%-95%
10	95%-100%

Another way in which we evaluated fatigue was by comparing their average maximum sprint speed (computed from the top ⅔ of plays) with their single maximum, relating to a player's ability to "dig deep" and realize their true maximum speed. Players with at least ten plays were used in this comparison to avoid issues with a small sample size. The difference between the average maximum and single maximum was computed to show the absolute difference from an individual play, and the proportion of plays within 95% of this average was also determined to better understand the distribution of speeds across plays.

Aside from exertion, we also performed simple analyses relating sprint speed directly with promotion. Conventional knowledge suggests that, while speed is a beneficial trait for players, it ultimately is not a very strong indicator of performance and promotion overall. A very fast player who struggles with bat-to-ball skills will be hampered by that deficiency, while slow players with great hitting prowess nonetheless find success. By testing this hypothesis, we also verify that there are no confounding variables in our analysis of fatigue.

RESULTS & DISCUSSION

Overall, comparing the distributions of players who were and were not promoted led to very similar results, indicating that this was not a strong distinguishing factor for promotion. This was not a wholly unsurprising result, considering the known lack of a strong correlation between speed and promotion, with fatigue as a somewhat related metric. As a sport with so many different components needed for success beyond pure athleticism, including hitting and defense, this is just one small component in the bigger picture. Ultimately, it is important to note that, while this result shows the inability to predict promotion purely based on speed and exertion, the converse is also not necessarily true, with performance dictated by a myriad of factors.

Histogram plots of the distribution for each metric between the two groups of players are included in **Appendix A**, as well as in the attached code files. All of the data exhibit similar behaviors, and the main source of distinction exists due to the much larger sample size of players who were not promoted. One important consideration with this comparison is that the groups of players should be divided more carefully in subsequent analysis. For example, a young talent and an aging veteran could be placed into the same group if promotion is the only point of comparison, but the actual situations of these players differ greatly. The anonymization of player information in this data set did not allow for such demographic divisions, but this is an approach that an organization could implement in practice.

Another interpretation of these results is to question whether any effect is a causation or correlation between exertion and promotion. The fact that physical exertion was not a predictor of the professional pipeline could indicate that organizations have not yet implemented load management as a point of evaluation, and so the benefits of accounting for this factor have yet to be realized. While it is still unlikely for this to be some magical formula, including load management and recovery as tangible metrics to evaluate prospects can potentially lead to new forms of player development that improve the career longevity of athletes.

One area of improvement for this specific methodology lies in our quantification of physical exertion. Our fatigue metric had its strengths but still needs refinement in terms of the fatigue factor values and quantitative efficiency. In future explorations, we may look to refine these aspects of the metric in addition to creating better coordination between its existence and the other metrics we valued in this study. Additionally, we had a smaller scope, so the metric we rendered may work differently under varying conditions. Still, the use of an RPE-esque system could be valuable in trying to quantify fatigue.

CONCLUSION

As technological advancements such as Hawkeye position tracking have been introduced in professional baseball, the increasing amounts of unexplored data provide a unique opportunity to combine the traditional approaches to sports strategy with the computational power of analytics. Among so many potential applications of this data, one of the most prominent fields is the consideration of player fatigue and load management. Developing an understanding of how to best foster player recovery could provide a competitive edge to an organization by maximizing the performance of their players.

This project analyzed fatigue on a subset of plays, infield ground balls, measuring it with a focus on its relationship to player promotion in a farm system. While the results did not immediately show any strong correlations, it is important to recognize that much work remains in this field. Infield ground balls represent only a fraction of player exertion, and finding a correlation to something else like player health would be equally beneficial for an organization. One interesting avenue to continue similar analyses could involve studying the relationship between player fatigue and injuries, applying a biomedical approach to the data. The constraints of the competition and data set limited the scope of our project, but this nonetheless serves as a preliminary foundation for future work.

Works Cited

Berg, S. (2024, July 12). *What doctors wish patients knew about Tommy John surgery*. American Medical Association. https://www.ama-assn.org/delivering-care/public-health/what-doctors-wish-patients-knew-about-tommy-john-surgery

Luff, C. (2024, June 30). *How to increase running stamina*. Verywell Fit. https://www.verywellfit.com/how-to-run-without-getting-tired-or-breathing-heavy-2911281

Petriello, M. (2023, February 21). *Trea Turner shoots up New Speed Leaderboard*. MLB.com. https://www.mlb.com/news/2018-update-to-sprint-speed-boosts-trea-turner-c272810784

Rogers, J. (2024, May 15). *How load management is becoming MLB's next big thing*. ESPN. https://www.espn.com/mlb/story/_/id/40145890/mlb-load-management-next-big-thing-rock-baseball

Statcast Sprint Speed Leaderboard. baseballsavant.com. (n.d.). https://baseballsavant.mlb.com/leaderboard/sprint_speed

APPENDIX A: FIGURES

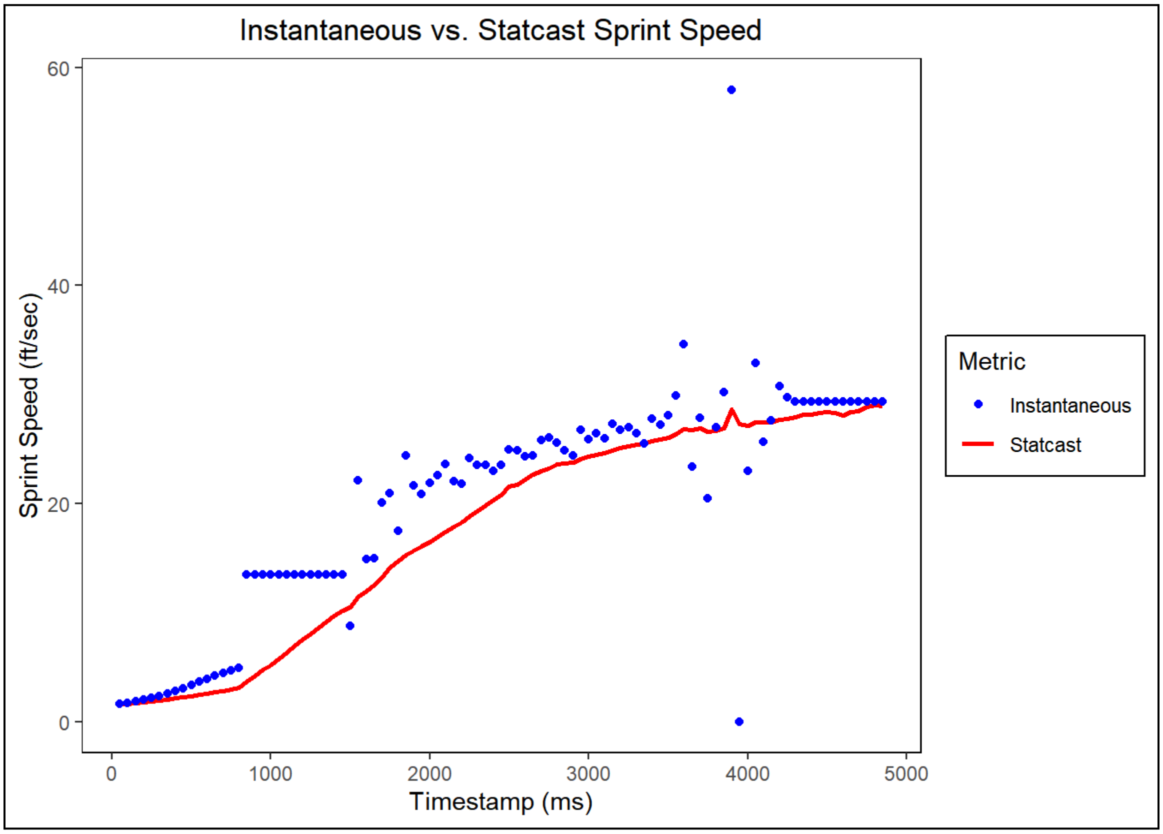

Figure 1. When plotting the instantaneous and Statcast sprint speeds for a given play, the large fluctuations and jumps in instantaneous speed (around 1000 and 4000 ms) motivate the use of Statcast sprint speed, reducing the impact of each individual point.

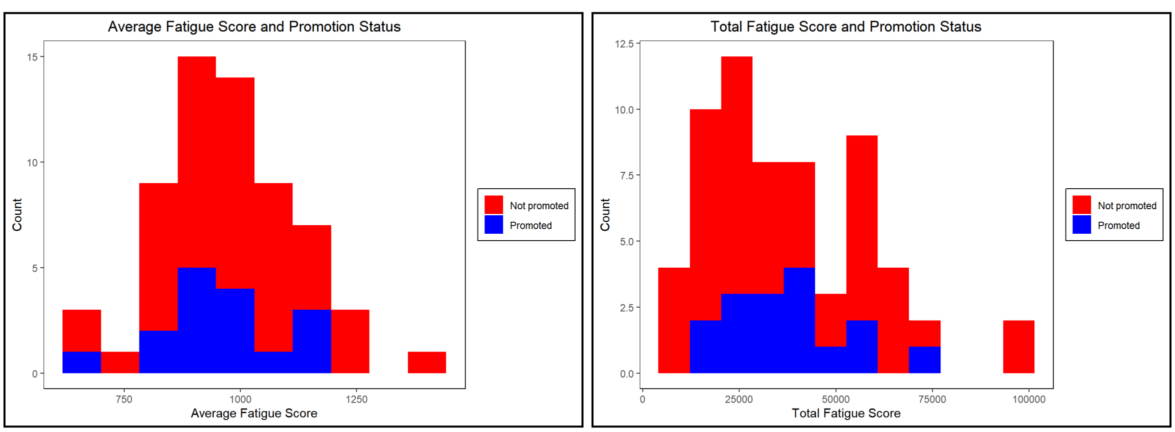

Figure 2. The distributions of average fatigue score (left) and total fatigue score (right) do not differ greatly between players who are (blue) and are not (red) promoted. Both distributions are shaped similarly, with the only major difference from the smaller sample size of promotions.

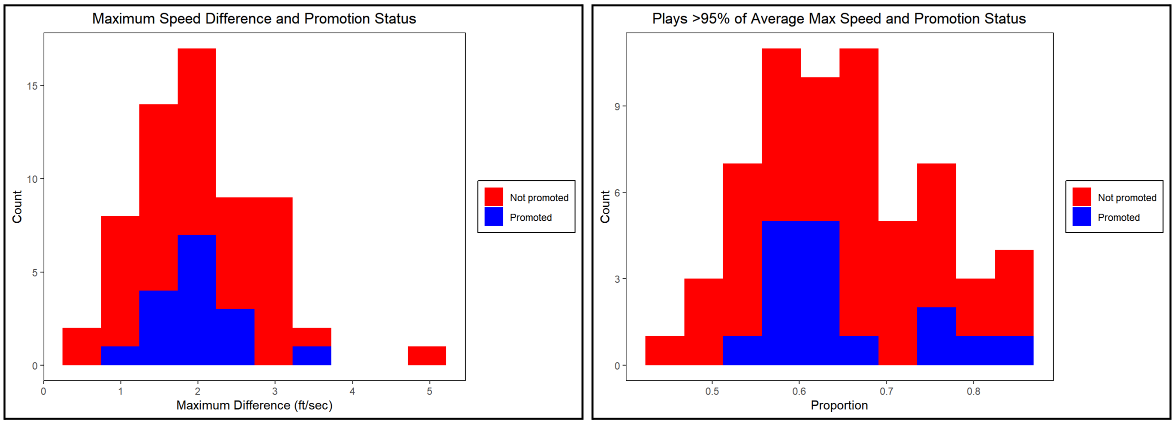

Figure 3. The distributions of maximum speed difference (left) and proportion of plays within 95% of maximum (right) do not differ greatly between players who are (blue) and are not (red) promoted. The shape of promoted and non-promoted distributions are very similar, appearing proportional to the size of each group.

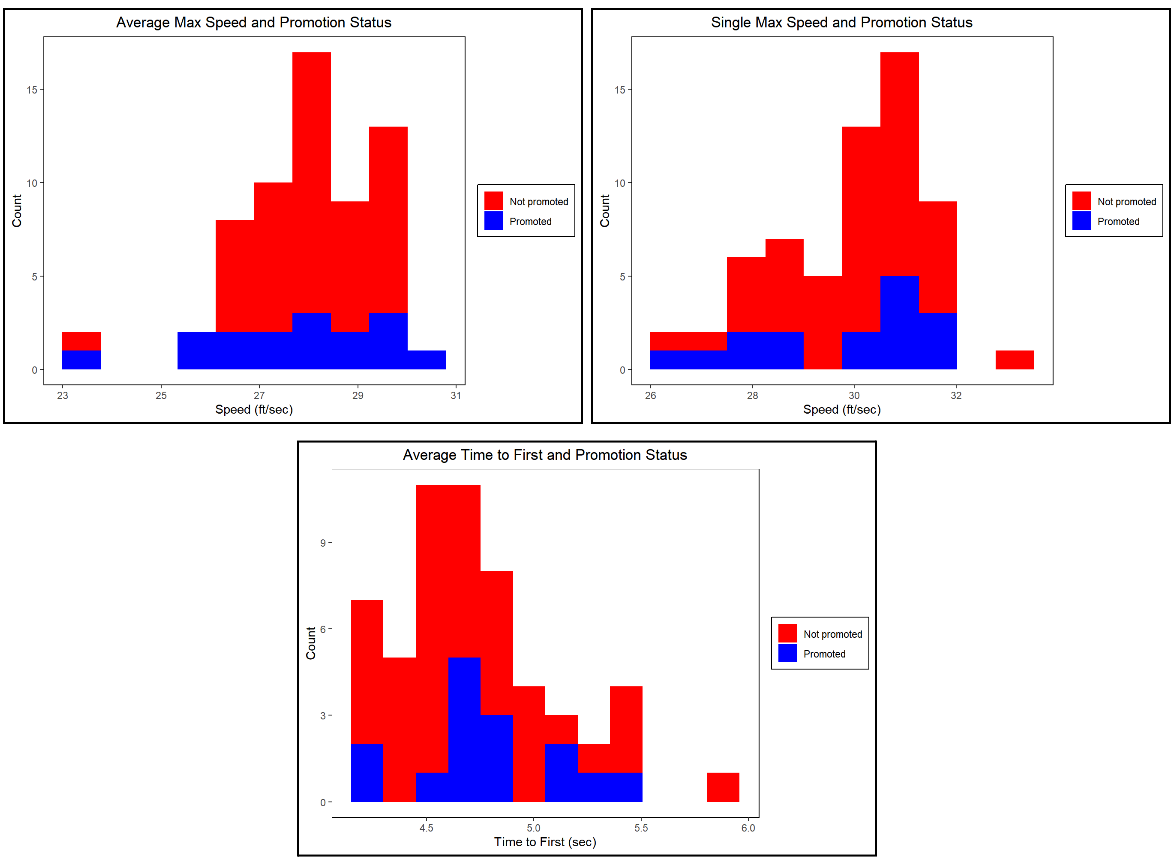

Figure 4. The distributions of average maximum speed (top left), single maximum speed (top right), and average time-to-first (bottom) do not differ greatly between players who are (blue) and are not (red) promoted. While there is a greater difference in the time-to-first plot, it is not significant enough to identify any conclusive relationship.

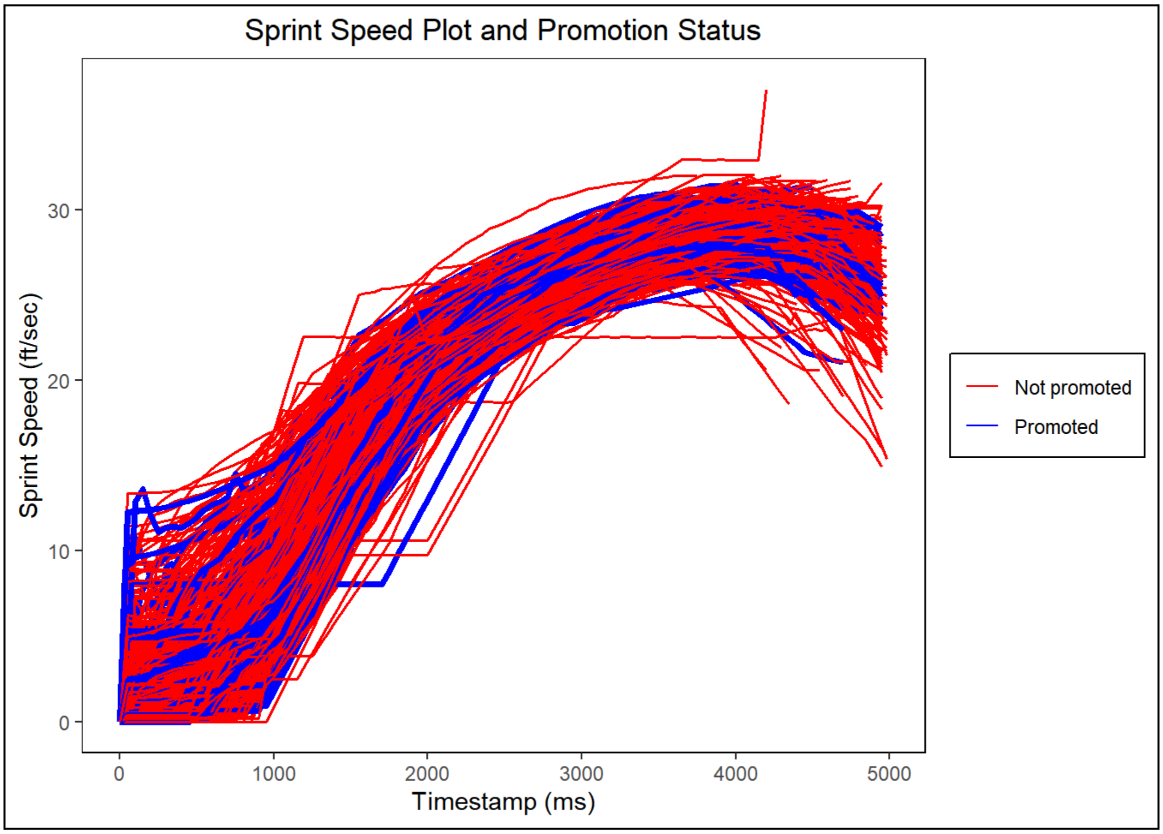

Figure 5. The sprint speed plots for selected plays (time-to-first < 4.5 sec) were graphed as a function of time. The general shape of the curve is largely the same for this set of plays, and there is no apparent difference in the positioning of promoted players (blue) compared to the overall trend.